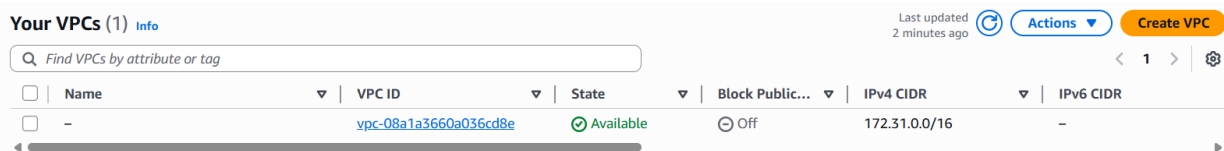


# VPC : VIRTUAL PRIVATE CLOUD

1. Login to AWS Management console .

2.Create VPC.

Go to VPC → Your VPC → Click Create VPC.



3. Click VPC only.

Enter.

Name tag – optional(test – vpc) → IPv4 CIDR(10.0.0.0/16) → Leave other option(default) → Click Create VPC.

## VPC settings

### Resources to create

Create only the VPC resource or the VPC and other networking resources.

☒ VPC only

☐ VPC and more

### Name tag - optional

Creates a tag with a key of 'Name' and a value that you specify.

test-vpc

### IPv4 CIDR block

☒ IPv4 CIDR manual input

☐ IPAM-allocated IPv4 CIDR block

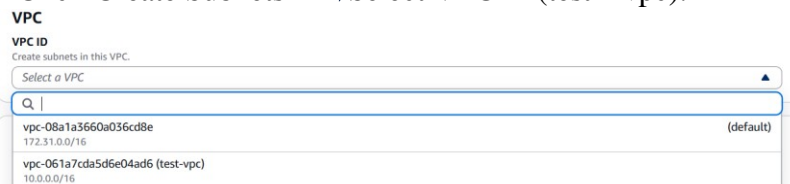
### IPv4 CIDR

10.0.0.0/16

CIDR block size must be between /16 and /28.

4. Create Subnets.

Go to Subnets → Click Create Subnets → Select VPC ID(test – vpc).



Create Public Subnet(10.0.1.0/24).

Create Private Subnet(10.0.2.0/24).

Select Availability Zones for high availability(Asia Pacific(Mumbai) / ap-south-1)

Enter.

Subnet name → Availability Zone → IPv4 Subnet CIDR block → Click create subnet.

Click Add new subnet(Similarly created public subnet)

**Subnet name**

Create a tag with a key of 'Name' and a value that you specify.

pub - sub1a

The name can be up to 256 characters long.

**Availability Zone** [Info](#)

Choose the zone in which your subnet will reside, or let Amazon choose one for you.

Asia Pacific (Mumbai) / ap-south-1a

**IPv4 VPC CIDR block** [Info](#)

Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

10.0.0.0/16

**IPv4 subnet CIDR block**

10.0.1.0/24

**Subnet name**

Create a tag with a key of 'Name' and a value that you specify.

pri - sub -1b

The name can be up to 256 characters long.

**Availability Zone** [Info](#)

Choose the zone in which your subnet will reside, or let Amazon choose one for you.

Asia Pacific (Mumbai) / ap-south-1a

**IPv4 VPC CIDR block** [Info](#)

Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

10.0.0.0/16

**IPv4 subnet CIDR block**

10.0.2.0/24

## 5. Create and Attach Internet Gateway (IGW).

Go to Internet Gateway.

Create Internet Gateway.

Attach the Internet Gateway to your VPC.

Enter.

IGW Name → Click Create IGW → Attach to VPC → Select(test - vpc) → Click Attach VPC.

**Internet gateway settings****Name tag**

Creates a tag with a key of 'Name' and a value that you specify.

test - vpc - igw

Select a VPC to communicate with

Attach to a VPC

**VPC**

Attach an internet gateway to a VPC to enable the VPC to communicate with the Internet.

**Available VPCs**

Attach the internet gateway to this VPC.

Select a VPC

vpc-061a7cda5d6e04ad6 - test-vpc

## 6. Create Route Table.

Create a Public Route Table and associate it with the Public Subnet.

Add a route to 0.0.0.0/0 via the Internet Gateway.

Enter.

Name and Tag optional → Select VPC(test - vpc) → Click create Route Table.

Select Route Table(Public) → Go to Subnet association &amp; Edit Subnet association →

Select public subnet → Click Save association.

Go to Routes → Click Edit routes → Click Add route → Click &amp; Select(0.0.0.0/0) →

Click &amp; Select Internet Gateway &amp; And also select(test - vpc) → Click Save changes.

**Route table settings****Name - optional**

Create a tag with a key of 'Name' and a value that you specify.

rt - pub

**VPC**

The VPC to use for this route table.

vpc-061a7cda5d6e04ad6 (test-vpc)

0.0.0.0/0

Add route

Internet Gateway

igw-05dc171ab53d6533c

Use: "igw-05dc171ab53d6533c"

igw-05dc171ab53d6533c (test - vpc - igw)

Keep the Private Route Table without internet access (default).

## 7. Launch an EC2 instance in Public subnet.

Create a new EC2 instance.

Enter.

Click "Launch Instance" → Provide a name for your instance (e.g. Win).

Choose Amazon Machine Image (AMI) → Select operating system (e.g. Windows).

Select Instance, choose t2.micro (Free Tier eligible).

Configure Network Settings:

Select the VPC you created earlier.

Choose Public Subnet.

Enable Auto-assign Public IP (so it can connect to internet).

Create/Choose Key Pair → For RDP login, create a new key pair or use existing one.

## 8. Configure Security Group:

Allow RDP(Port 22) → For remote login.

Allow HTTP (Port 80) → For web access.

(Optional) Allow HTTPS (Port 443) if needed.

Review and Launch → Verify all settings, then click Launch Instance.

Verify Instance Running → In EC2 Dashboard, check instance status (should be “Running”). Connect to Instance:

## 9. Inside the Network setting.

VPC - required [Info](#)

vpc-061a7cda5d6e04ad6 (test-vpc)  
10.0.0.0/16

Subnet [Info](#)

subnet-0e23f24d1e30b1af6 pub - sub1a  
VPC: vpc-061a7cda5d6e04ad6 Owner: 711387100611  
Availability Zone: ap-south-1a (aps1-az1) Zone type: Availability Zone  
IP addresses available: 251 CIDR: 10.0.1.0/24

Auto-assign public IP [Info](#)

Enable

Additional charges apply when outside of free tier allowance



Type [Info](#)

rdp

Protocol [Info](#)

TCP

Source type [Info](#)

Anywhere

Source [Info](#)

[Add CIDR, prefix list or security group](#)

0.0.0.0/0 X

▼ Security group rule 2 (TCP, 80, 0.0.0.0/0)

Type [Info](#)

HTTP

Protocol [Info](#)

TCP

Source type [Info](#)

Anywhere

Source [Info](#)

[Add CIDR, prefix list or security group](#)

0.0.0.0/0 X

## 9. Launching Instance(Public Subnet) In RDP.

Log in

Enter the Workspace's username and password.

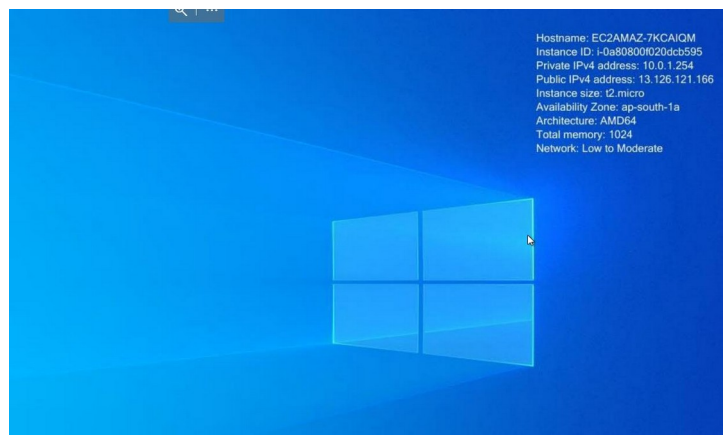
Username

Administrator

Password

Enter password

Connect Cancel



## Testing Phase: Compilation of Code (error detection)

No error was found.

## Implementation Phase: Final Output(no error)

A vpc with a public subnet was created, an Internet Gateway was attached, and an EC2 instance was launched successfully with internet connectivity verified.

